

SAN VALENTINO ALLA MUTA, Italy

WMO: 160080

Lat: 46.7619N

Lon: 10.5344E

Elev: 1461

StdP: 84.96

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.7	(c) -11.7	(d) -19.9	(e) 0.8	(f) -9.8	(g) -17.9	(h) 0.9	(i) -7.7	(j) 8.7	(k) -7.0	(l) 7.5	(m) -5.8	(n) 2.5	(o) 10	(p) 0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 24.9	(d) 14.3	(e) 23.1	(f) 13.9	(g) 21.5	(h) 13.5	(i) 16.6	(j) 21.3	(k) 15.7	(l) 19.9	(m) 14.9	(n) 19.1	(o) 2.5	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.1	12.8	18.0	14.2	12.1	17.2	13.4	11.4	16.5	52.5	21.5	49.5	20.0	47.0	19.0	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.0	(b) 5.8	(c) 5.1	(e) -16.0	(f) 27.9	(g) 2.8	(h) 2.1	(i) -18.0	(j) 29.4	(k) -19.7	(l) 30.6	(m) -21.3	(n) 31.7	(o) -23.3	(p) 33.2		
			DB	WB												
					-16.4	18.7	3.0	1.8	-18.6	19.9	-20.3	21.0	-22.0	22.0	-24.1	23.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-3.0	-2.0	2.0	5.6	9.7	13.7	15.4	15.0	11.5	7.6	2.3	-1.6		
	DBStd	7.40	3.80	4.19	3.96	3.57	3.11	3.50	2.99	2.84	3.03	3.52	3.90	3.92		
	HDD10.0	1878	404	336	248	138	43	8	2	2	19	88	231	359		
	HDD18.3	4374	663	569	507	383	267	143	99	108	206	332	481	617		
	CDD10.0	562	0	0	0	5	34	119	168	157	63	14	1	0		
	CDD18.3	15	0	0	0	0	0	3	7	5	0	0	0	0		
	CDH23.3	143	0	0	0	0	1	42	54	43	3	0	0	0		
	CDH26.7	16	0	0	0	0	0	5	5	6	1	0	0	0		
(14) Wind	WSAvg	2.0	2.0	2.2	2.1	2.3	2.1	2.0	2.0	1.8	2.0	2.0	2.0	2.1		
(15) Precipitation	PrecAvg	701	35	33	38	37	61	84	103	103	65	58	52	38		
	PrecMax	896	85	159	93	79	123	125	166	187	140	155	158	107		
	PrecMin	509	7	7	8	7	18	39	70	43	23	1	4	4		
	PrecStd	98	21	31	21	16	24	23	24	35	30	34	34	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	10.9	15.2	19.6	22.1	26.9	27.2	27.1	22.3	19.1	14.5	9.2		
		MCWB	3.0	3.4	5.5	9.5	11.9	14.1	15.8	15.1	12.7	10.1	8.2	3.2		
	2%	DB	5.2	7.8	12.5	16.3	19.6	24.3	24.8	24.2	20.4	16.2	11.0	6.5		
		MCWB	1.4	2.3	4.8	7.7	10.6	13.7	14.6	14.4	12.4	9.6	6.1	2.1		
	5%	DB	3.3	5.3	10.0	14.0	17.5	22.2	23.0	22.2	18.4	14.4	8.8	4.5		
		MCWB	0.4	0.8	3.6	6.5	9.8	13.4	14.2	14.1	11.9	9.1	5.0	1.2		
	10%	DB	1.7	3.2	7.7	11.8	15.7	20.1	21.2	20.3	16.4	12.6	6.8	3.0		
		MCWB	-0.6	-0.2	2.7	5.6	9.2	12.7	13.8	13.7	11.2	8.8	4.3	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.4	7.0	10.5	13.9	17.2	18.5	17.5	15.5	12.7	9.4	4.4		
		MCDB	7.0	9.2	13.0	16.2	18.6	21.9	23.2	22.9	19.7	15.7	12.5	6.4		
	2%	WB	2.1	2.7	5.5	8.9	12.3	15.9	16.8	16.2	14.1	11.3	7.1	3.1		
		MCDB	4.3	6.9	10.7	14.2	16.6	20.7	21.6	20.4	17.8	14.0	9.5	5.1		
	5%	WB	0.8	1.3	4.3	7.5	11.2	14.7	15.8	15.4	13.1	10.3	5.7	1.8		
		MCDB	2.6	4.5	8.7	12.2	15.3	19.4	20.1	19.4	16.0	12.8	7.6	3.8		
	10%	WB	-0.2	0.2	3.1	6.4	10.2	13.7	15.0	14.7	12.3	9.2	4.7	0.7		
		MCDB	1.4	2.6	6.8	10.3	13.8	18.2	19.0	18.5	15.1	11.8	6.4	2.6		
(35) Mean Daily Temperature Range	MDBR		7.4	8.3	9.1	9.1	8.7	9.5	9.6	8.9	8.3	7.7	6.3	6.5		
	5% DB	MCDBR	9.0	11.1	12.7	13.0	12.2	12.7	12.5	12.0	11.4	10.8	9.6	8.3		
		MCWBR	6.0	6.7	6.4	5.7	5.1	4.8	4.7	4.5	4.8	5.3	5.5	5.3		
	5% WB	MCDBR	8.0	10.1	11.4	11.3	9.9	10.8	10.4	9.5	9.3	8.6	7.7	7.4		
		MCWBR	5.7	6.5	6.3	5.7	5.1	5.2	5.2	4.7	4.8	5.0	5.0	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.269	0.316	0.366	0.381	0.388	0.384	0.377	0.341	0.315	0.276	0.243		
	taud		2.330	2.259	2.199	2.189	2.243	2.275	2.289	2.329	2.413	2.456	2.455	2.380		
	Ebn at Noon		869	906	904	886	883	876	874	867	872	846	821	829		
	Edh at Noon		72	96	121	135	133	130	126	117	98	80	65	61		
(44) All-Sky Solar Radiation	RadAvg		1.56	2.51	3.71	4.47	4.88	5.56	5.60	4.90	3.76	2.54	1.54	1.29		
	RadStd		0.17	0.29	0.44	0.61	0.53	0.45	0.44	0.46	0.39	0.32	0.25	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A
(47) Regional (107 neighbors)		+0.34	N/A	N/A	+0.63	+0.40	+0.34	N/A	N/A	+60	+15

Nomenclature: See separate page